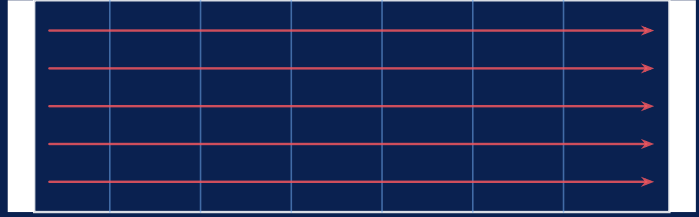


4



CHAPTER

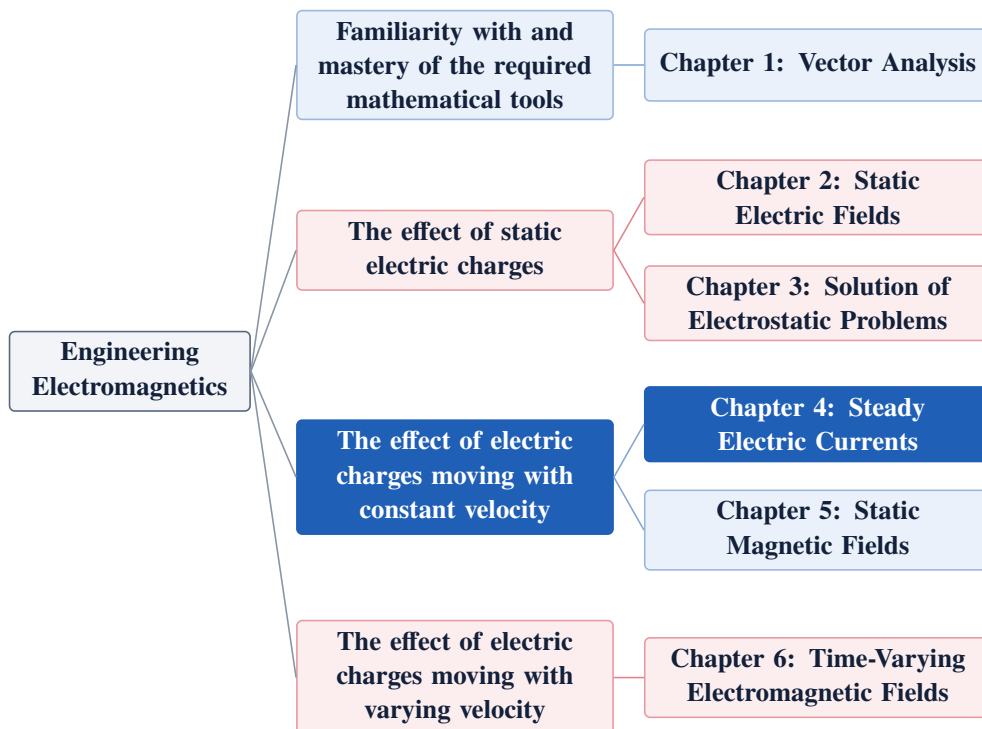
Steady Electric Currents

جریان‌های الکتریکی دائم

In this chapter

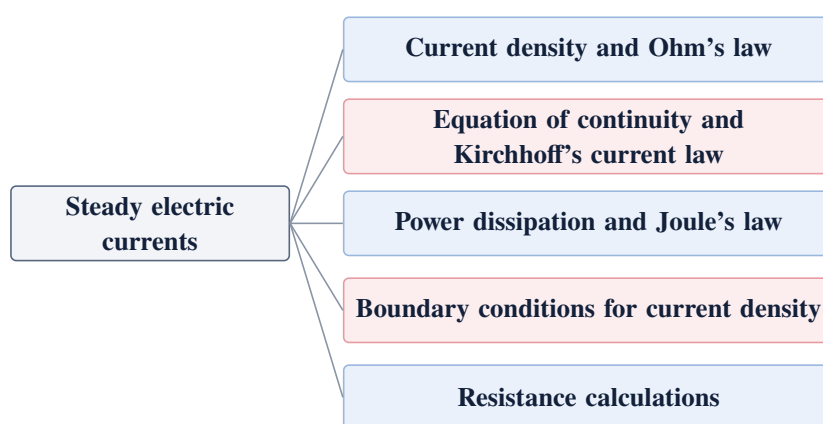
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4.1 Course Outline



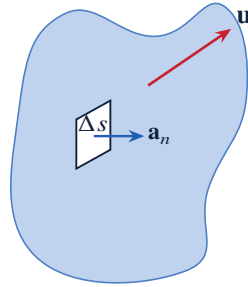
4.2 Steady Electric Currents

- In the previous chapters we dealt with electrostatic problems due to electric charges at rest.
 - In this chapter we turn to moving electric charges, which produce electric current.
- Electric currents due to the motion of free charges are divided into three kinds
 - **Conduction currents:** produced in conductors and semiconductors by the drift motion of electrons or holes.
 - **Electrolytic currents:** due to the motion of positive and negative ions
 - **Convection currents:** due to the motion of electrons or ions in vacuum
- In this chapter we concentrate on conduction currents, which are governed by Ohm's law.



- What is the concept of the **volume current density**, and what is the relation for the **convection current density**?
- What is the relation for the **conduction current density**?
- What is the point form of **Ohm's law**?
- What is the **equation of continuity**, and how is **Kirchhoff's current law** derived from it?
- What is **Joule's law**, which gives the power dissipated by a current flowing in a conducting body?

4.3 Current Density and Ohm's Law



- Consider the steady motion of charge carriers, each with charge q , moving with velocity $\mathbf{u}$ across a small surface element Δs
- If N is the number of charge carriers per unit volume, the charge passing through the surface Δs in the time Δt is

$$\Delta Q = Nq\mathbf{u} \cdot \mathbf{a}_n \Delta s \Delta t \quad (\text{C})$$

- Since current is the time rate of change of charge, we have

$$\Delta I = \frac{\Delta Q}{\Delta t} = Nq\mathbf{u} \cdot \mathbf{a}_n \Delta s = Nq\mathbf{u} \cdot \Delta \mathbf{s} \quad (\text{A})$$

- We define the volume current density, in amperes per square metre, as follows

$$\mathbf{J} = Nq\mathbf{u} \quad (\text{A/m}^2)$$

⚡ Important result | Convection current density

$$\mathbf{J} = \rho\mathbf{u} \quad (\text{A/m}^2)$$

- Therefore

$$\Delta I = \mathbf{J} \cdot \Delta \mathbf{s}$$

- Thus the total current flowing through an arbitrary surface S is

⚡ Important result

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} \quad (\text{A})$$

- Conduction currents are the result of the drift motion of charge carriers under the influence of an applied electric field.
- For most conducting materials the average drift velocity is directly proportional to the electric field intensity.

– For metallic conductors we have

$$\mathbf{u} = -\mu_e \mathbf{E} \quad (\text{m/s})$$

- μ_e : electron mobility, in $\text{m}^2/\text{V} \cdot \text{s}$

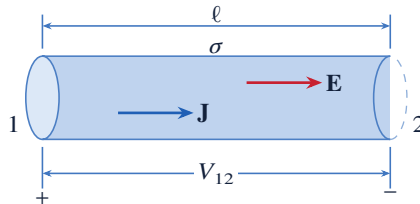
$$\mathbf{J} = -\rho_e \mu_e \mathbf{E}$$

- $\rho_e = -Ne$: the volume charge density of the drifting electrons, a negative quantity

⚡ Important result | Point form of Ohm's law

$$\mathbf{J} = \sigma \mathbf{E} \quad (\text{A/m}^2)$$

- $\sigma = -\rho_e \mu_e$: a fundamental parameter of the medium, called the conductivity
- The unit of conductivity is $\text{A/V} \cdot \text{m}$, or siemens per metre. The reciprocal of conductivity is called resistivity, in $\Omega \cdot \text{m}$.
- Obtaining the relation between the voltage and the current of a piece of homogeneous material of conductivity σ , length ℓ and uniform cross-section S from the point form of Ohm's law



$$\left. \begin{aligned} V_{12} = E\ell &\Rightarrow E = \frac{V_{12}}{\ell} \\ I = \int_S \mathbf{J} \cdot d\mathbf{s} = JS &\Rightarrow J = \frac{I}{S} \end{aligned} \right\} \frac{I}{S} = \sigma \frac{V_{12}}{\ell}$$

$$V_{12} = \left(\frac{\ell}{\sigma S} \right) I = RI$$

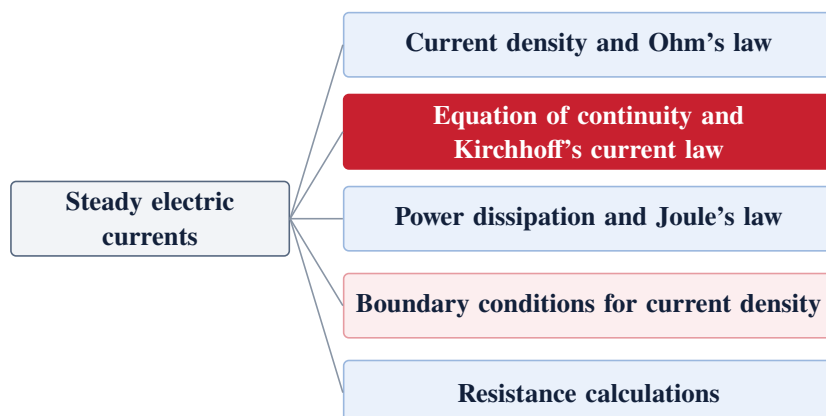
⚡ Important result

$$R = \frac{\ell}{\sigma S} \quad (\Omega)$$

$$G = \frac{1}{R} = \sigma \frac{S}{\ell} \quad (\text{S})$$

- G : conductance, the reciprocal of resistance, in mhos or siemens

4.4 Equation of Continuity and Kirchhoff's Current Law



- The principle of conservation of charge is one of the fundamental postulates of physics.
 - Electric charge can be neither created nor destroyed.
- Consider an arbitrary volume V bounded by the surface S , containing a net charge Q . If a net current I flows out of the region through the surface, the charge in the volume must decrease at a rate equal to the current. Conversely, if a net current flows into the region through the surface, the charge in the volume must increase at a rate equal to the current.
- The current leaving the region is the total outward flux of the current density vector through the surface S

$$I = \oint_S \mathbf{J} \cdot d\mathbf{s} = -\frac{dQ}{dt} = -\frac{d}{dt} \int_V \rho \, dv$$

$$\int_V \nabla \cdot \mathbf{J} \, dv = - \int_V \frac{\partial \rho}{\partial t} \, dv$$

- Since the above equation must hold regardless of the choice of V

⚡ Important result | Equation of continuity

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} \quad (\text{A/m}^3)$$

- If there are no time variations

$$\nabla \cdot \mathbf{J} = 0$$

⚡ Important result | Kirchhoff's current law

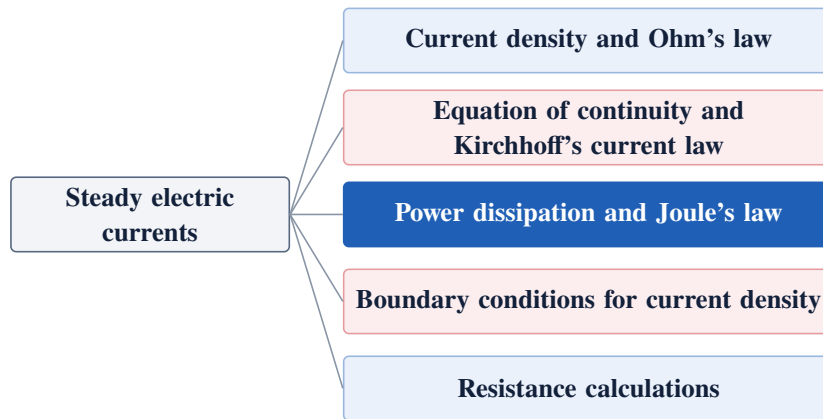
$$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$$

- ▶ We said earlier that charges placed inside a conductor move to its surface, so that at equilibrium $\rho = 0$ inside the conductor. We now want to prove this statement

$$\left. \begin{array}{l} \sigma \nabla \cdot \mathbf{E} = -\frac{\partial \rho}{\partial t} \\ \nabla \cdot \mathbf{E} = \rho / \epsilon \end{array} \right\} \quad \frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon} \rho = 0 \quad \Rightarrow \quad \rho = \rho_0 e^{-(\sigma/\epsilon)t} \quad (\text{C/m}^3)$$

- ▶ ρ_0 : initial charge density
- ▶ The above equation states that the charge density at a given point decreases exponentially with time.
- ▶ The initial density decays to $1/e$, or 36.8%, of its value in a time equal to $\tau = \epsilon/\sigma$.
 - For a good conductor such as copper, $\tau = 1.52 \times 10^{-19}$ s.
 - For a good insulator this time may be hours or days.

4.5 Power Dissipation and Joule's Law



- ▶ Under the influence of an electric field, the free electrons in a conductor move and collide with the atoms of the crystal lattice. Energy is thus transferred from the electric field to the vibrating atoms.
- ▶ The work done by the electric field in moving a charge q through the distance $\Delta \ell$ is

$$\Delta W = \mathbf{F} \cdot \Delta \ell = q\mathbf{E} \cdot \Delta \ell$$

- ▶ This work corresponds to the power

$$p = \lim_{\Delta t \rightarrow 0} \frac{\Delta w}{\Delta t} = q\mathbf{E} \cdot \mathbf{u}$$

- ▶ $\mathbf{u}$: the velocity of the charge carriers

$$dp = \rho dv \mathbf{E} \cdot \mathbf{u}$$

$$dP = \mathbf{E} \cdot \mathbf{J} dv$$

- For a given volume V the total power converted into heat is

⚡ Important result | Joule's law

$$P = \int_V \mathbf{E} \cdot \mathbf{J} dv \quad (\text{W})$$

- In a conductor of constant cross-section we have

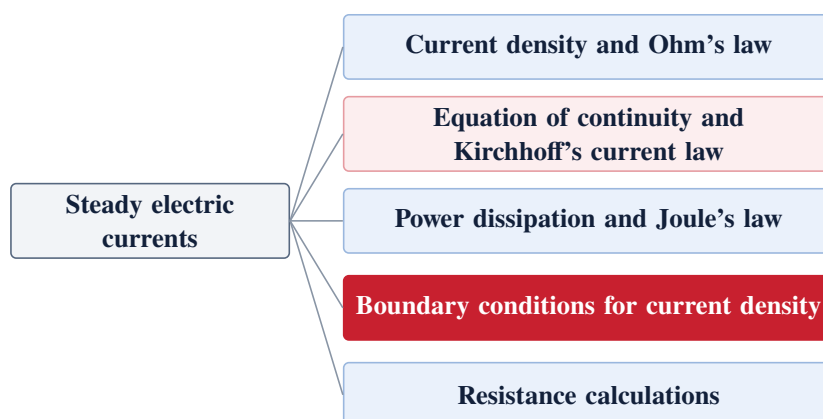
$$P = \int_L E d\ell \int_S J ds = VI$$

⚡ Important result

$$P = I^2 R \quad (\text{W})$$

- which is the familiar relation for ohmic power and gives the heat dissipated in the resistance R .

4.6 Boundary Conditions for Current Density



- When a current crosses the interface between two media of different conductivities obliquely, the current density vector changes in both direction and magnitude.
- The governing equations for the steady current density are

Governing Equations for Steady Current Density

Differential Form	Integral Form
$\nabla \cdot \mathbf{J} = 0$	$\oint_S \mathbf{J} \cdot d\mathbf{s} = 0$
$\nabla \times \left(\frac{\mathbf{J}}{\sigma} \right) = 0$	$\oint_C \frac{1}{\sigma} \mathbf{J} \cdot d\boldsymbol{\ell} = 0$

- The boundary conditions are as follows

⚡ Important result

$$\nabla \cdot \mathbf{J} = 0 \quad \Rightarrow \quad J_{1n} = J_{2n} \quad (\text{A/m}^2)$$

$$\nabla \times (\mathbf{J}/\sigma) = 0 \quad \Rightarrow \quad \frac{J_{1t}}{J_{2t}} = \frac{\sigma_1}{\sigma_2}$$

- When a steady current crosses the boundary between two different lossy dielectrics

$$J_{1n} = J_{2n} \rightarrow \sigma_1 E_{1n} = \sigma_2 E_{2n}$$

$$D_{1n} - D_{2n} = \rho_s \rightarrow \epsilon_1 E_{1n} - \epsilon_2 E_{2n} = \rho_s$$

$$\rho_s = \left(\epsilon_1 \frac{\sigma_2}{\sigma_1} - \epsilon_2 \right) E_{2n} = \left(\epsilon_1 - \epsilon_2 \frac{\sigma_1}{\sigma_2} \right) E_{1n}$$

- $E_{1n} = \mathbf{a}_{n2} \cdot \mathbf{E}_1$ and $E_{2n} = \mathbf{a}_{n2} \cdot \mathbf{E}_2$
- If medium 2 is a much better conductor than medium 1

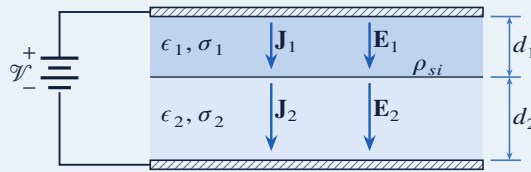
$$\rho_s = \epsilon_1 E_{1n} = D_{1n}$$

- which is the boundary condition of a conductor.

Example 4-1

A potential difference $\mathcal{V}$ is applied to a parallel-plate capacitor of area S . The space between the conducting plates is filled with two lossy dielectric materials of thicknesses d_1 and d_2 , permittivities ϵ_1 and ϵ_2 , and conductivities σ_1 and σ_2 . Find

- (a) the electric field intensity in the two media
- (b) the current density in the two media
- (c) the surface charge densities on the conducting plates and at the interface of the two dielectrics



➤ Solution.

- (a)

$$J_1 = J_2 \quad \Rightarrow \quad \begin{aligned} \mathcal{V} &= E_1 d_1 + E_2 d_2 \\ \sigma_1 E_1 &= \sigma_2 E_2 \end{aligned}$$

$$E_1 = \frac{\sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2} \quad E_2 = \frac{\sigma_1 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

- (b)

$$J_1 = J_2 = \sigma_1 E_1 = \sigma_2 E_2 = \frac{\sigma_1 \sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

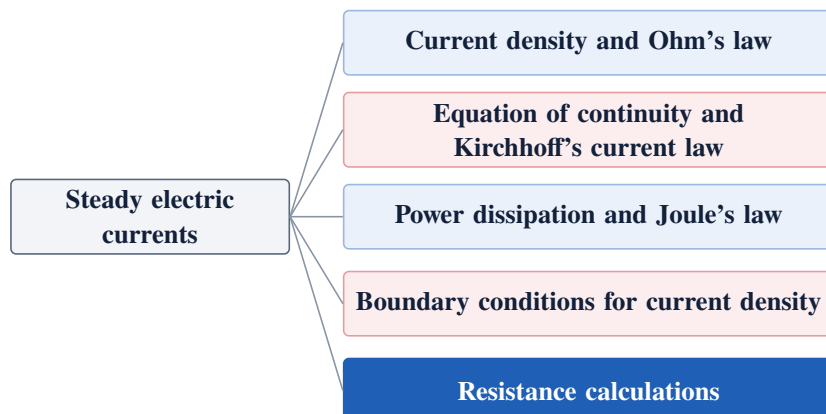
- (c)

$$\mathbf{a}_n \cdot \mathbf{E}_1 = \frac{\rho_{s1}}{\epsilon_1} \quad \Rightarrow \quad \rho_{s1} = \epsilon_1 E_1 = \frac{\epsilon_1 \sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

$$\mathbf{a}_n \cdot \mathbf{E}_2 = \frac{\rho_{s2}}{\epsilon_2} \quad \Rightarrow \quad \rho_{s2} = -\epsilon_2 E_2 = -\frac{\epsilon_2 \sigma_1 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2}$$

$$\rho_{si} = \left(\epsilon_1 - \epsilon_2 \frac{\sigma_1}{\sigma_2} \right) E_{1n} \quad \Rightarrow \quad \begin{aligned} \rho_{si} &= \left(\epsilon_2 \frac{\sigma_1}{\sigma_2} - \epsilon_1 \right) \frac{\sigma_2 \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2} \\ &= \frac{(\epsilon_2 \sigma_1 - \epsilon_1 \sigma_2) \mathcal{V}}{\sigma_2 d_1 + \sigma_1 d_2} \quad (\text{C/m}^2) \end{aligned}$$

4.7 Resistance Calculations



- How is the **leakage resistance** of capacitive structures computed?
- How is the resistance of various structures computed?
- The capacitance between two conductors separated by a dielectric medium is

$$C = \frac{Q}{V} = \frac{\oint_S \mathbf{D} \cdot d\mathbf{s}}{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}} = \frac{\oint_S \epsilon \mathbf{E} \cdot d\mathbf{s}}{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}}$$

- The surface integral is taken over a surface enclosing the positive conductor, and the line integral from the negative conductor to the positive conductor.
- If the dielectric medium is lossy, a current flows from the positive to the negative conductor. The resistance between the conductors (the leakage resistance) is

$$R = \frac{V}{I} = \frac{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}}{\oint_S \mathbf{J} \cdot d\mathbf{s}} = \frac{-\int_L \mathbf{E} \cdot d\boldsymbol{\ell}}{\oint_S \sigma \mathbf{E} \cdot d\mathbf{s}}$$

- If the permittivity and the conductivity have the same space dependence ($\epsilon(x, y, z) = k\sigma(x, y, z)$), or if the dielectric medium is homogeneous (ϵ and σ independent of the space coordinates)

⚡ Important result

$$RC = \frac{C}{G} = \frac{\epsilon}{\sigma}$$

Example 4-2

Find the leakage resistance per unit length

- (a) between the inner and outer conductors of a coaxial cable with inner-conductor radius a , outer-conductor radius b and a dielectric of conductivity σ
- (b) of a two-wire transmission line consisting of two wires of radius a separated by D , in a medium of conductivity σ

➤ Solution.

- (a) The capacitance per unit length of a coaxial cable

$$C_1 = \frac{2\pi\epsilon}{\ln(b/a)} \quad (\text{F/m})$$

- The leakage resistance per unit length

$$R_1 = \frac{\epsilon}{\sigma} \left(\frac{1}{C_1} \right) = \frac{1}{2\pi\sigma} \ln \left(\frac{b}{a} \right) \quad (\Omega \cdot \text{m})$$

- (b) The capacitance per unit length of a two-wire transmission line

$$C'_1 = \frac{\pi\epsilon}{\cosh^{-1} \left(\frac{D}{2a} \right)} \quad (\text{F/m})$$

- The leakage resistance per unit length

$$\begin{aligned} R'_1 &= \frac{\epsilon}{\sigma} \left(\frac{1}{C'_1} \right) = \frac{1}{\pi\sigma} \cosh^{-1} \left(\frac{D}{2a} \right) \\ &= \frac{1}{\pi\sigma} \ln \left[\frac{D}{2a} + \sqrt{\left(\frac{D}{2a} \right)^2 - 1} \right] \quad (\Omega \cdot \text{m}) \end{aligned}$$

- In certain situations electrostatic and steady-current problems are not exactly analogous.
- **First method** for computing the resistance of a piece of conducting material between two surfaces or terminals

- Choosing an appropriate coordinate system
- Applying a potential difference V_0 between the two terminals
- Finding the electric field intensity (if the material is homogeneous, by solving Laplace's equation for the electric potential and then computing the electric field intensity)
- Finding the total current

$$I = \int_S \mathbf{J} \cdot d\mathbf{s} = \int_S \sigma \mathbf{E} \cdot d\mathbf{s}$$

- S : the cross-section through which the current flows
- The resistance is V_0/I

- **Second method** for computing the resistance of a piece of conducting material between two surfaces or terminals (when J can easily be determined from I)

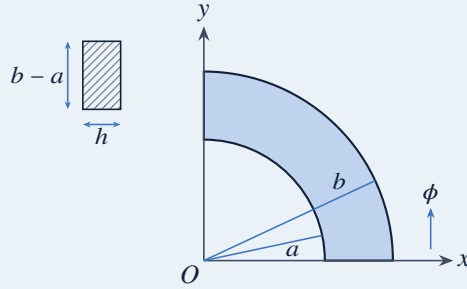
- Choosing an appropriate coordinate system
- Passing a current I between the two terminals
- Finding J from I
- Computing the electric field intensity $\mathbf{E} = \mathbf{J}/\sigma$ and the potential difference V_0

$$V_0 = - \int \mathbf{E} \cdot d\boldsymbol{\ell}$$

- The integration is carried out from the terminal at the lower potential to the terminal at the higher potential.
- The resistance is V_0/I

Example 4-3

Consider a conducting material of uniform thickness h and conductivity σ in the shape shown in the figure below. Find the resistance between its two end faces.



► Solution.

► Choosing cylindrical coordinates

$$V = 0 \quad \text{at} \quad \phi = 0, \quad V = V_0 \quad \text{at} \quad \phi = \pi/2$$

$$\frac{d^2 V}{d\phi^2} = 0 \quad \Rightarrow \quad V = c_1 \phi + c_2 \quad \Rightarrow \quad V = \frac{2V_0}{\pi} \phi$$

$$\begin{aligned} \mathbf{J} &= \sigma \mathbf{E} = -\sigma \nabla V \\ &= -\mathbf{a}_\phi \sigma \frac{\partial V}{r \partial \phi} = -\mathbf{a}_\phi \frac{2\sigma V_0}{\pi r} \end{aligned}$$

$$\begin{aligned} I &= \int_S \mathbf{J} \cdot d\mathbf{s} = \int_0^h \int_a^b \left(-\mathbf{a}_\phi \frac{2\sigma V_0}{\pi r} \right) \cdot (-\mathbf{a}_\phi dr dz) \\ &= \frac{2\sigma h V_0}{\pi} \ln \frac{b}{a} \end{aligned}$$

$$R = \frac{V_0}{I} = \frac{\pi}{2\sigma h \ln(b/a)}$$